5.) Here you the have below the [illegible]
accordingly to [illegible] [illegible] according to [illegible]

$X = 1 \in E$ $\quad \frac{E_y}{E_x} = \frac{b_y}{b_x}$ $\quad \frac{E_y}{E_x} = \frac{b_y}{b_x}$ $\quad \frac{E_y}{E_x} = \frac{b_y}{b_x}$ $\quad \frac{E_y}{E_x} = \frac{b_y}{b_x}$

$\frac{E_y}{E_x} = b_0 \times \cos \quad \frac{E_y}{E_x} = \frac{b_y}{b_x c}$

[illegible] [illegible] $\quad$ X [illegible] $\quad$ [illegible]

$\frac{\partial E}{\partial \theta} = \frac{\partial}{\partial \theta}\left(\frac{E_y}{E_x}\right) = \left[\frac{\partial}{\partial \theta}\left(\frac{b_y}{b_x}\right) = \frac{\partial}{\partial \theta}\left(\frac{b_y}{b_x}\right)\right]$

$\lambda = \frac{E_y}{E_x} = \frac{b_y}{b_x} = \frac{1}{1+\varepsilon(1-\cos\theta)}$ $\quad$ and $\varepsilon = \frac{E_y}{E_x c}$ [illegible]

[illegible] [illegible] [illegible]

$\eta(\theta) = \frac{E - E_y}{E} = \frac{E - \frac{E_y}{1+\varepsilon(1-\cos\theta)}}{E} = \varepsilon\left[1 - \frac{1}{1+\varepsilon(1-\cos\theta)}\right]$

$\eta(\theta) = E_y\left[1 - \frac{1}{1+\varepsilon(1-\cos\theta)}\right]$ $\quad$ [illegible] [illegible] [illegible]

$\frac{d\eta}{d(\cos\theta)} = E_y\left[\frac{(1+\varepsilon(1-\cos\theta))^2(-\varepsilon)}{\quad}\right]$

$\frac{d\eta}{d(\cos\theta)} = \frac{\varepsilon E_y}{(1+\varepsilon(1-\cos\theta))^2}$ $\quad$ (I)

now from $E = N$: [illegible]

$\frac{d\tau}{d\theta} = \frac{d\tau}{d\theta}\frac{d(\cos\theta)}{d(\cos\theta)} = \frac{\tau_0 c}{\lambda}\left[\frac{1}{1+\varepsilon(1-\cos\theta)}\right]\left[\frac{\quad}{1+\varepsilon(1-\cos\theta)}\right]$ [illegible]

[illegible] $\quad$ [illegible] [illegible] [illegible]

$\frac{d\tau}{d(\cos\theta)} = \tau_0 c\left[\frac{1}{1+\varepsilon(1-\cos\theta)}\right]\left[\frac{1}{1+\varepsilon(1-\cos\theta)} + (1-\cos\theta)\frac{\quad}{\quad}\right]$

$\frac{d\tau}{d(\cos\theta)} = \tau_0 c\left[\frac{1}{1+\varepsilon(1-\cos\theta)}\right]^2\left[\frac{\quad}{1+\varepsilon(1-\cos\theta)}\right]$

now: $\frac{d\tau}{d(\cos\theta)} = \tau_0 c\,(\text{I})$

$\frac{d\tau}{d(\cos\theta)} = \frac{d(\cos\theta)}{d\eta} = \frac{d\tau}{d\eta}$

$\frac{d\tau}{d\eta} = \frac{\tau_0 c}{\varepsilon E_y}\left[\frac{\quad}{1+\varepsilon(1-\cos\theta)} + \varepsilon(1-\cos\theta) + \varepsilon^2(1-\cos\theta)^2\right]$

now still: $\eta(\theta) = E_y\left[1 - \frac{1}{1+\varepsilon(1-\cos\theta)}\right]$

if [illegible] [illegible] $\cos\theta = \xi$

so

$\frac{d\tau}{d\eta} = \frac{\tau_0 c}{\varepsilon E_y}\left[\frac{\quad}{1+\varepsilon(1-\xi)} + \varepsilon(1-\xi) + \varepsilon^2(1-\xi)^2\right]$

$\eta = E_y\left[1 - \frac{1}{1+\varepsilon(1-\xi)}\right]$

now choose: $\frac{d\tau}{d\eta}$ from ξ & ε are [illegible] of η:

$\frac{\eta + E_y}{E_y} = \frac{1}{1+\varepsilon(1-\xi)}$

$\varepsilon(1-\xi) = \frac{E_y}{\eta - E_y}$

$\xi = 1 - \frac{1}{\varepsilon}\left(\frac{E_y}{\eta - E_y} - 1\right) = 1 - \frac{1}{\varepsilon}\left(\frac{\eta}{\eta - E_y} - 1\right)$

so $\frac{d\tau}{d\eta} = \frac{\tau_0 c}{\varepsilon E_y}\left[\varepsilon_1 - \eta - \frac{E_y}{\eta - E_y} - \frac{1}{\varepsilon^2}\left(\frac{\eta}{\eta - E_y} - 1\right)^2\right]$

now we need to normalize: the [illegible] function

normalization constant: A:

$A = \int_{\xi_1}^{\xi_2}\left(\frac{d\tau}{d\eta}\right)d\eta$ $\quad$ [illegible] $\eta_{max} = \eta = E_y$ [illegible] $= E_y\left[1 - \frac{1}{1+\varepsilon E_y}\right]$ $\quad$ σc

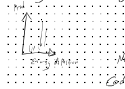
we can solve A numerically in [illegible]

so then: $P(\eta) = \frac{1}{A}\frac{d\tau}{d\eta}$

so then we can start the make [illegible]

process using the $P(\eta)$: as our normalized

energy distribution probability function!



Please review
next Bizet [illegible] [illegible];
Cooler, so [illegible] [illegible] [illegible]
[illegible] the HW.